Programming in HTML5 with JavaScript and CSS3 Chapter 16: Offline web applications

Q1. When you need more advanced features such as true asynchronous support, indexing for faster searching, or transactions for offline web storage, which of the following you might use?

A. localStorage
B. sessionStorage
C. Web SQL
D. Indexeddb

Answer: C, D

[Page 581]

Q2. Which one of the following offline web storage is an object database providing the power of indexing and transactions without the need to set up a formal relational structure?

A. localStorage
B. sessionStorage
C. Web SQL
D. Indexeddb

Answer: D [Page 581]

Q3. Which one the following storage is suitable for storing files such as images, text files, XML, or even movies?

A. HTTP cache
B. Filesystem API
C. Web SQL
D. Indexeddb

Answer: B [Page 581]

Q4. Which one the following can you use to make an entire website offline-friendly with very little effort?

A. HTTP cache
B. Filesystem API
C. Web SQL
D. Indexeddb

Answer: A [Page 581]

Q5. Which one provides all the features of a relational database on the client side?

A. HTTP Cache
B. Web SQL
C. IndexedDB
D. Filesystem API

Answer: B
[Page 582, 591
IndexedDB is object-based database]

Q6. Which one provides object-based database on the client side?

A. Web storage
B. Web SQL
C. IndexedDB
D. Filesystem

API Answer: C
[Page 581]

Q7. Which one is ideal for storage of files like text file, xml files or even media files on the client side?

A. Web storage
B. Web SQL
C. IndexedDB
D. Filesystem

API Answer: D
[Page 581]

Q8. Which database standard the most current implementation of Web SQL is built on?

A. SQL Server
B. MySql
C. SQLite
D. Oracle

Answer: C
[Page 582
SQLite is popular in-memory embedded database]

Q9. When you attempt to open a database openDtatabase method of the Web SQL API, what will happen if the database you are trying to open does not exist?

A. the openDatabase method will return 'undefined'
B. a database will be created automatically
C. an exception will be thrown
D. a prompt will be opened asking to enter database

name Answer: B
[Page 582]

Q10. Which one is the complete definition of the openDatabase method of Web SQL API?

A. openDatabase (name, version)
B. openDatabase (name, version, displayName)
C. openDatabase (name, version, displayName, creationCallback)
D. openDatabase (name, version, displayName, estimatedSize,

creationCallback) Answer: D
[Page 583]

Q11. When is the creationCallback passed in the openDatabase method invoked?

A. Always, if already exists or if the database does not yet exist and is being created, the callback will be invoked

B. If the database exists, the callback will be invoked
C. If the database does not yet exist and is being created, the callback will be invoked
D. If the database version is changed, the callback will be invoked

Answer: C
[Page 583]

Q12. Which code block tries to open a database Library database of size 1MB if exists, otherwise it should be created and a callback function is invoked?

A. var db = openDatabase('Library', '1.0', 'My library', 1 * 1024 * 1024); db.oncreate = (db) =>{/*code*/}
B. var db = openDatabase('Library', '1.0', 'My library', 1 * 1024 * 1024); db.onsuccess = (db) =>{/*code*/}
C. var db = openDatabase('Library', '1.0', 'My library', 1 * 1024 * 1024); db.open = (db) =>{/*code*/}
D. var db = openDatabase('Library', '1.0', 'My library', 1 * 1024 * 1024,(db) =>{/*code*/}); Answer: D

[Page 584
If database does not exist it will be created and creationCallback is called]

Q13. How do you close a database connection with Web SQL?

A. Call the close method on database object.
B. Call the transaction on the database object, inside the method call the commit method on the transaction object
C. Set database object to null
D. You don't have to close the connection manually, it is automatically handled

Answer: D
[Page 583]

Q14. What is the typical default database size in Web SQL?

A. 1 MB
B. 5 MB
C. 10 MB
D. Unlimited

Answer: B
[Page 583]

Q15. How do apply modifications such as adding or dropping a table to an existing database with the Web SQL?

A. Use changeVersion method
B. Use transaction method
C. Use creationCallback in openDatabase method
D. Use executeSql method

Answer: A
[Page 583]

Q16. Which code fragment drops an existing table, adds a new table to the existing database?

A. var db = openDatabase('mydb', '1.0','my db', 1*1024*1024); db.transaction((tx) =>{
   tx.executeSql("DROP TABLE tblold");

```
        tx.executeSql("CREATE TABLE tblnew (I'd INTEGER, code TEXT) ") ;
    });
```

B. var db = openDatabase('mydb', '1.0','my db', 1*1024*1024); db.changeVersion('1.0','2.0',(tx) =>{

```
        tx.executeSql("DROP TABLE tblold") ;
        tx.executeSql("CREATE TABLE tblnew (I'd INTEGER, code TEXT) ") ;
    }, (err) =>{/*error*/}, () =>{/*done*/}) ;
```

C. var db = openDatabase('mydb', '2.0','my db',

```
        1*1024*1024,(db)=>{ db.transaction((tx) =>{
        tx.executeSql("DROP TABLE tblold") ;
      tx.executeSql("CREATE TABLE tblnew (I'd INTEGER, code TEXT) ") ;
    }) ;
```

D. var db = openDatabase('mydb', '2.0','my db', 1*1024*1024); db.transaction('1.0','2.0',(tx) =>{

```
      a. tx.executeSql("DROP TABLE tblold") ;
      b. tx.executeSql("CREATE TABLE tblnew (I'd INTEGER, code TEXT) ") ;
    }, (err) =>{/*error*/}, () =>{/*done*/});
```

Answer: B
[Page 583
The best option is changeVersion call and pass a callback. Do the modifications in the callback.]

Q17. The name of the database you pass to openDatabase method is case-sensitive.

A. True
B. False

Answer: A
[Page 583]

Q18. Which code block correctly creates a database and create a table in it?

A. var db = openDatabase('mydb', '1.0', 'my first database', 2 * 1024 * 1024); db.executeSql('CREATE TABLE IF NOT EXISTS T1 (id unique, text)');
B. var db = openDatabase('mydb', '1.0', 'my first database', 2 * 1024 * 1024); db.transaction('CREATE TABLE IF NOT EXISTS T1 (id unique, text)');
C. var db = openDatabase('mydb', '1.0', 'my first database', 2 * 1024 * 1024); db.transaction(function (tx) {
  tx.executeSql('CREATE TABLE IF NOT EXISTS T1 (id unique, text)'); });
D. var db = openDatabase('mydb', '1.0', 'my first database', 2 * 1024 * 1024); db.create(function (tx) {
  tx.executeSql('CREATE TABLE IF NOT EXISTS T1 (id unique, text)');
  });

Answer: C
[Page 584]

Q19. Which method of the Database object in Web SQL provides a transaction to execute SQL statements allowing both read and write commands?

A. transaction
B. readTransaction
C. writeTrasnsaction
D. executeSql

Answer: A
[Page 584]

Q20. What is the required parameter of the transaction method of the Database object in Web SQL?

A. A callback function
B. A success callback function
C. An error callback function
D. A Database object

Answer: A
[Page 585]

Q21. What does the callback function of the Database object's transaction method accept as parameter?

A. A Database object
B. A transaction object
C. A SQL statement
D. Array of objects

Answer: B
[Page 585]

Q22. With Web SQL, Schema migration support is available by using the _____method.

A. migrated()
B. changeVersion()
C. transaction()
D. executeSql()

Answer: B
[Page 589]

Q23. In Web SQL, what does the openDatabase method do?

A. Opens a database if exists and returns the opened database. If the database does not exist returns null
B. Opens a database and returns it if it exists otherwise throws exception
C. Opens a database, if it does not exist it create one and returns it
D. Opens a database if exists and returns the opened database. If the database does not exist does nothing

Answer: C
[Page 588]

Q24. Which one is not a valid call to open a Web SQL database?

A. var db = openDatabase('Library');
B. var db = openDatabase('Library', '1.0', 'My library');
C. var db = openDatabase('Library', '1.0', 'My library', 5 * 1024 * 1024);
D. var db = openDatabase('Library', '1.0', 'My library', 5 * 1024 * 1024,
function(e){ alert('New db created');
});

Answer: A
[Page 583]

Q25. You are working with Web SQL. Your code is like below

```
1. var db = openDatabase('Library', '1.0', 'My library', 5 * 1024 * 1024);
2. function migrateDB(t) {
3.     t.executeSql("CREATE TABLE IF NOT EXISTS authors(" +
4.         "id INTEGER PRIMARY KEY AUTOINCREMENT, name TEXT)");
5. }
6. function onError(error) {
7.     alert("Error code: " + error.code + " Message: " + error.message);
8. }
9. function onSuccess() {
10.     alert("Migration complete!");
11. }
13.
```

You want change version of the database. Which code would you add at line 13?

A. db.changeVersion('1.0' , '2.0', onSuccess, onError, migrateDB);
B. db.changeVersion('1.0' , '2.0', onError, onSuccess, migrateDB);
C. db.changeVersion('1.0' , '2.0', migrateDB, onSuccess, onError);
D. db.changeVersion('1.0' , '2.0', migrateDB, onError, onSuccess);

Answer: D
[Page 584]

Q26. To insert a row in a table in Web SQL database which method of the Web SQL database object do you use?

A. transaction
B. executeSql
C. insert
D. add

Answer: A
[Page 585
Don't confuse with executeSql which is a method of transaction object. Transaction is the method of Database object.

```
var db = openDatabase(……);
db.transaction((tx)=>{ //callback function which accepts transaction parameter
        tx.executeSql("…….");
});
```

Q27. Which one following is the required parameter of the transaction method of the Web SQL database object?

A. Callback to execute command
B. Callback method to invoke if an error occurs while the transaction is being processed
C. Callback method to invoke if all statements successfully execute within the transaction
D. All of the above

Answer: A
[Page 585]

Q28. What commands does the readTransaction method of the Web SQL database object support?

A. Read commands
B. Insert commands
C. Update commands
D. Delete commands

Answer: A
[Page 585]

Q29. Which one is **not** true about IndexedDB?

A. IndexedDB use tables with defined schema
B. IndexedDB uses object stores, which are key/value storage areas
C. IndexedDB a key/value database in which values can range from simple strings to complex object structures
D. IndexedDB maintains indexes over the records it stores for fast retrieval

Answer: A
[Page 589]

Q30. When upgradeneeded event is trigger if you call to open an IndexedDB database?

A. if the requested database does not exist
B. if the version requested does not match the current version of the existing database
C. if the database exceeds the size allowed in browser settings
D. if version is not supplied with the database name

Answer: A, B
[Page 591]

Q31. Which code block correctly creates an object store in an IndexedDB database?

A.
```
var indexedDB = window.indexedDB;
var openRequest = indexedDB.open('Library', 1);
openRequest.onsuccess = function(response) {
        var db = openRequest.result;
                db. createObjectStore("authors", { keypath: 'id', autoIncrement: true });
};
```
B.
```
var indexedDB = window.indexedDB;
var openRequest = indexedDB.open('Library', 1);
openRequest.onupgradeneeded = function(response) {
        response.currentTarget.result.createObjectStore("authors", { keypath:
'id', autoIncrement: true });
};
```
C.
```
var indexedDB = window.indexedDB;
var openRequest = indexedDB.open('Library', 1);
openRequest. onupgradeneeded = function(response) {
        var db = openRequest.result;
        db. createObjectStore("authors", { keypath: 'id', autoIncrement: true });
};
```
D.
```
var indexedDB = window.indexedDB;
var openRequest = indexedDB.open('Library', 1);
openRequest.onsuccess = function(response) {
        response.currentTarget.result.createObjectStore("authors", { keypath: 'id',
autoIncrement: true });
};
```

Answer: B
[Page 591]

Q32. Which of the following methods should you use when storing files that are relatively unimportant and can be removed if the browser is low in available space?

A. window.requestFileSystem(LOW, 5 * 1024 * 1024, getDirectory, handleError);
B. window.requestFileSystem(PERSISTENT, 5 * 1024 * 1024, getDirectory, handleError);
C. window.requestFileSystem(SIMPLE, 5 * 1024 * 1024, getDirectory, handleError);
D. window.requestFileSystem(TEMPORARY, 5 * 1024 * 1024, getDirectory, handleError); Answer: D

[Page 601]

Q33. Which of the following allows FileWriter to append data to the end of the file?

A. fileWriter.seek(fileWriter.length);
B. fileWriter.seek(fileWriter.end);
C. fileWriter.moveLast()
D. fileWriter.moveToEnd();

Answer: A

[Page 603]

Q34. Which of the following removes a directory that contains existing files?

A. directoryEntry.remove(directoryRemoved, handleError);
B. directoryEntry.removeRecursively(directoryRemoved, handleError);
C. directoryEntry.removeAll(directoryRemoved, handleError);
D. directoryEntry.remove(directoryRemoved, ALL, handleError) Answer: B

[Page 606]

Q35. Which of the following is a valid statement for the first line of a manifest file?

A. CACHE
B. CACHE MANIFEST
C. CACHE-MANIFEST
D. CACHE-ALWAYS Answer: C

{page 608]

Q36. Which event is fired after an existing cache has been updated with new resources?

A. downloading
B. cached
C. updateReady
D. completed

Answer: C [Page 610]

Q37. You are working with indexedDB. Your code is like below.

```
1. var indexedDB = window.indexedDB;
2. var openRequest = indexedDB.open('Library', 1);
3. var db;
4. openRequest.onsuccess = function(evt) {
5.
6. };
```

You want to store opened database in db variable. Which code statement is correct?

A. db = openRequest.result;
B. db = openRequest
C. db = evt.result;
D. db =
evt.target.result; Answer: D
[Page 596]

Q38. Consider code
var indexedDB = window.indexedDB = window.indexedDB || window.mozIndexedDB || window.webkitIndexedDB || window.msIndexedDB;
What is the type of indexedDB variable?
A. indexedDB object
B. IDB
C. IDBFactory object
D. Simple JavaScript object
Answer: C

Q39. How does open method IDBFactory object work?
A. It is synchronous process, it opens the target database and returns the reference
B. It is asynchronous process, it begins an asynchronous process to open database and fires onsuccess event when database is ready to use.
C. It is asynchronous process, it begins an asynchronous process to open database and invokes the callback any supplied to notify when database is ready.
D. None of the above
Answer: B
[Page 591]

Q40. How does indexedDB store data?
A. indexedDB stores data in tables where Each table contains a set of columns, each of which has a name and a data type.
B. IndexedDB uses spaces called object stores, which are key/value storage areas.
C. IndexedDB stores data in key/value pairs and both key and value can be JSON objects
D. IndexedDB stores data in key/value pairs and both key and value are strings
Answer: B
[Page 591]

Q41. What happens if supplied version to indexedDB open method does not match with the current version?
A. An exception is thrown
B. Version of the database is changed to supplied version
C. If version supplied is higher than the current one, version of the database is changed otherwise an exception is thrown
D. An onupgradeneeded event that will be triggered
Answer: D
[Page 591]